# SEASIDERS

# OSMOTIC POWER INTEGRATED DESALINATION SYSTEM (THEME 1)

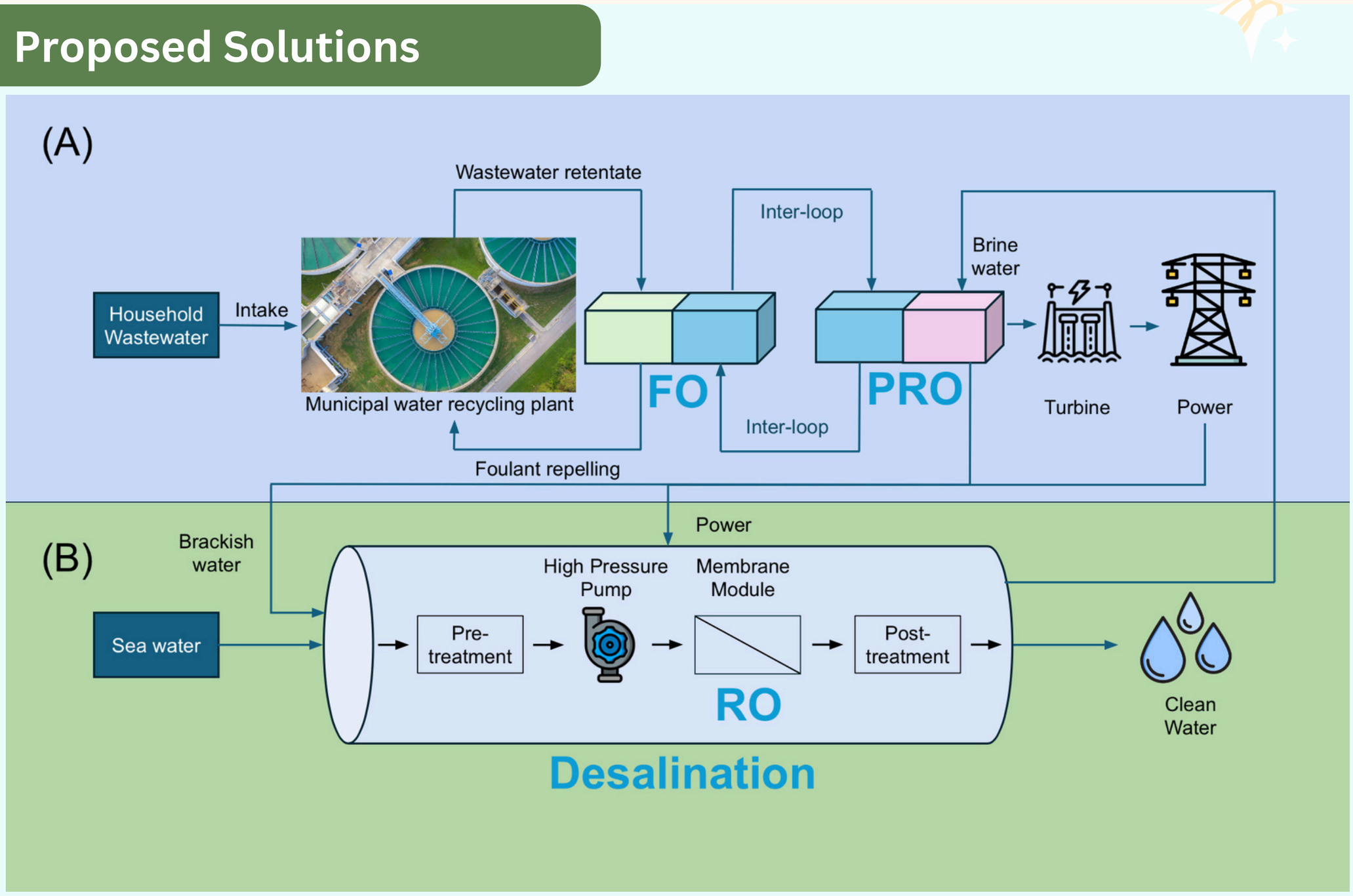
Team Member:
Chai Jin Xie
Lam Nguyen Ngan Ha
Natalie Tanjaya
Si Poh Yi

## Focused Problems

**More than 1 in 4 people worldwide lack access to safe drinking water services, and the number of urban residents without safe drinking water is expected to double by 2050. To address this issue, many countries have been actively constructing desalination plants as their primary clean water source. The global desalination plant capacity has doubled since the last decade and is projected to double again by 2050, benefiting over 500 million people, particularly in Middle East countries, supplying more than half of the nation's drinking and industrial water. However, these plants face common challenges. Disposal of brine byproduct that contains highly concentrated salt particles often leads to increased salinity levels in coastal waters, which deteriorates the water quality. Desalination plants are also notorious for their tremendous energy consumption. In water-scarce Singapore alone, every cubic meter of water requires a whopping 3.5kWh energy from non-renewable fossil fuels and imported energy.**

**To tackle these, osmotic power, as a prevalent renewable energy, shall be integrated with desalination due to their mutually-supporting production methods. But before that, we need to overcome the existing efficiency limitations due to the relatively low osmotic gradient between seawater and freshwater, which pose great challenges as they result in energy wastage during power generation and reduce economical viability. It is crucial to balance clean water provision and power sufficiency by systematically integrating energy and water production to achieve sustainable impacts. Therefore, we propose SEAsiders, an innovative system that integrates osmotic power and desalination, featuring enhanced membrane efficiency, minimized impacts on natural water bodies, and utilizes energy recovery to ensure the desalination plant's self-resilience, ultimately achieving greater energy and water security and improving welfare for everyone.**

## Proposed Solutions




## Estimated System Efficiency

**Our integrated system aims to achieve the following:**

**Reduce power consumption from current 3.5kWh/m3 to 1.8kWh/m3. An approximately 50% saving.**

**Extend membrane lifespan from existing 3-5 years to 6-8 years**

**Energy Recovery > 50%**

**Zero Brine Disposal**

**Cost Reduction > 30%**

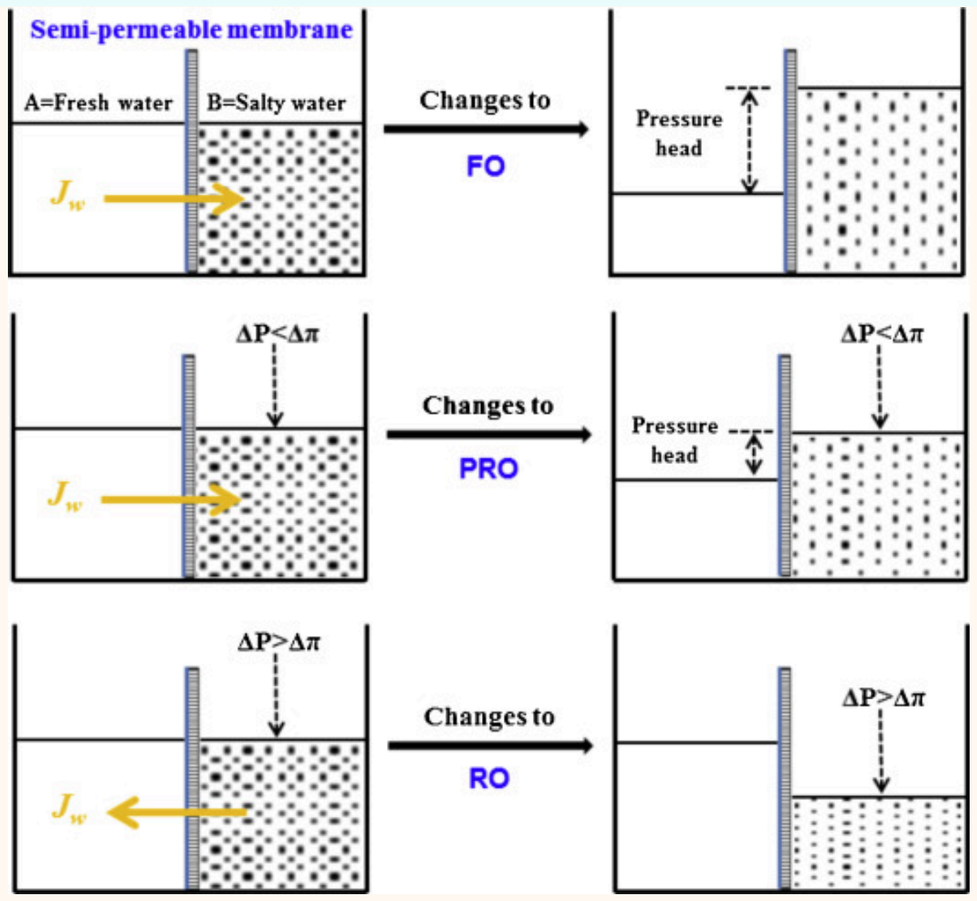



Figure 1. Three different osmotic processes

## (A) Osmotic Power

**FO-PRO hybrid technology is implemented in the osmotic power plant design:**

**1) Intake Requirements:**

**Draw solution with high salinity and feed solution with low salinity.**

**2) Forward Osmosis (FO) Process:**

**Wastewater retentate serves as feed solution here and through natural diffusion driven by the osmotic potential difference (Figure 1), water molecules pass through the membrane, removing foulants to ensure the cleanliness of solution in the inter-loop.**

**3) Inter-loop:**

**The inter-loop solution, which contains slightly higher salinity than the wastewater retentate, serves as the draw solution in the FO process to create clean and diluted feed solution for PRO process and protect the PRO membrane from fouling.**

**4) Pressure Retarded Osmosis (PRO) Process:**

**The concentrated brine water from the desalination facility acts as the draw solution and pumped into the pressure chamber (Appendix 1). The osmotic gradient between two solutions generates osmotic pressure, driving the flow of water molecules through the semipermeable membrane.**

**5) Power Extraction:**

**During the PRO process , the draw side pressure increases and is harnessed to drive a turbine or other energy conversion devices(Appendix 2). The turbine converts the pressure into mechanical energy, which is then further converted into electrical power.**

**6) Integration with Desalination Facility:**

**Generated electricity can be effectively utilized to power high-pressure pumps for desalination, thereby reducing the overall energy consumption of the desalination facility. The brackish water created from the PRO process can be directed to the desalination facility for RO processing as a source to desalinate.**

## (B) Desalination Plant

**Reverse Osmosis (RO) technologies are utilized in desalination facilities to convert seawater and brackish water into fresh water by eliminating dissolved salts and impurities. Here's the breakdown of the processes involved:**

**1) Pretreatment: This stage involves removing large particles, suspended solids, and organic matter from saline water. Its purpose is to safeguard the RO membranes from fouling and extend their lifespan.**

**2) High-pressure Pumping: Pressurized pumps are employed to control the pressure of the saline water. This high pressure creates the driving force necessary for water to pass through the RO membranes. It's important to note that this stage consumes a significant amount of energy due to the required pressure levels.**

**3) Reverse Osmosis (RO): The pressurized saline water is directed through semi-permeable membranes(Figure 1). These membranes contain extremely small pores that allow water molecules to pass through while blocking larger salt and impurity particles. As a result, fresh water is separated from the saline water, leaving behind a concentrated brine stream.**

**4) Post-treatment: The desalinated water produced by the RO process undergoes additional treatment processes to enhance its quality and meet the requirements for drinkable water. These processes may include remineralization, pH adjustment, disinfection, and carbon filtration.**

## Key Component : Semipermeable Membrane

**Semipermeable membranes are the workhorses behind desalination and osmotic power generation. They act as selective filters, allowing only water molecules or specific ions to pass through, while blocking dissolved salts or impurities. In desalination, they separate freshwater from seawater, and in osmotic power, they exploit the natural movement of water from low to high salinity to generate pressure, which is further converted into energy.**

**However, the specific requirements differ for each application:**

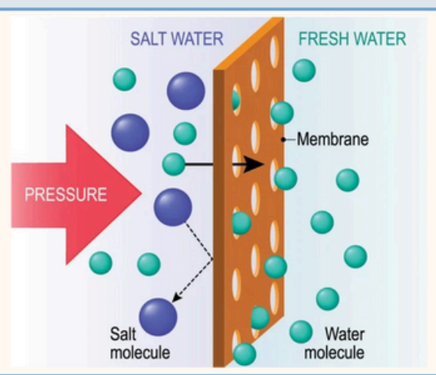
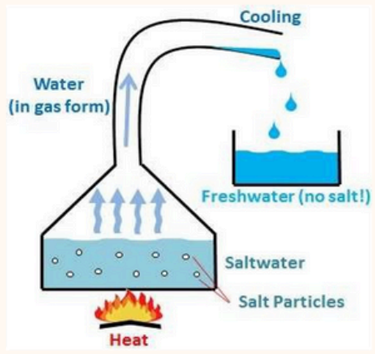
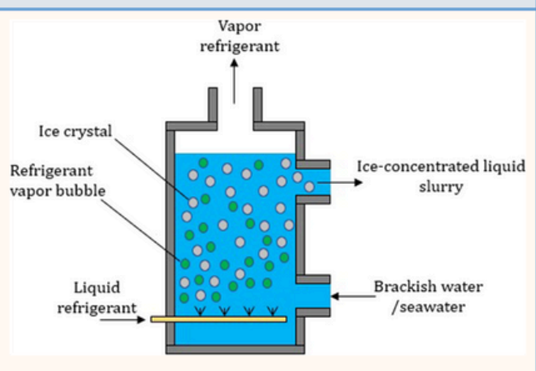
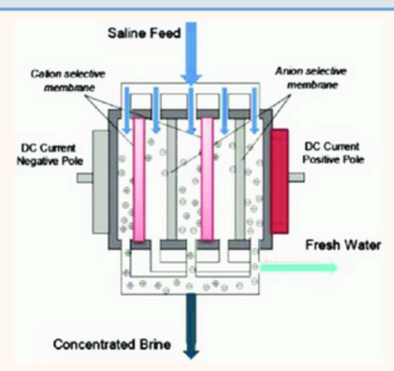
**1) FO membranes prioritize resistance to organic fouling for easier cleaning on the active layer.**

**2) PRO membranes balance roughness and strength to minimize fouling on the support layer and withstand pressure.**

**3) RO membranes focus on high pressure resistance and anti-fouling properties on the feed-facing surface.**

| Features | Nanocomposite membrane |
|---|---|
| Filtering function | • The gap between two nano particles is small, efficiency in filtering water molecule and other molecules.<br>• This allows for precise separation of desired molecules while rejecting unwanted ones |
| Advantages | 1. Nanoparticles significantly enhance the strength, lifespan.<br>2. They resist contaminant build-up, require less cleaning, and have better chemical and thermal stability.<br>3. Lower energy consumption and offer long-term cost savings through reduced maintenance.<br>4. Their superior performance and durability make them the clear choice for various applications. |

# Comparison to Existing Solutions

| Features | Reverse Osmotic (RO) | Thermal Distillation (TD) | Crystallization | Electrodialysis (ED) |
|---|---|---|---|---|
| **Description** | RO involves forcing seawater through a semi-permeable membrane that blocks salts and other impurities while allowing fresh water to pass through. | TD is a thermal desalination process where seawater is heated and then flashed into steam in multiple stages. | Freeze desalination involves partially freezing seawater or brackish water, forming pure ice crystals that exclude salts, which concentrate in the remaining brine. The ice crystals are then separated from the brine and melted to produce fresh water | ED use electric fields to move salts through a series of membranes, separating fresh water from brine. |
| **Illustration** |  |  |  |  |
| **Advantages** | • High efficiency in salt removal, especially for high salinity seawater<br>• Modular and scalable.<br>• Lower energy consumption compared to thermal methods. | • Reliable and can handle high salinity feed water.<br>• Long operational life with proper maintenance. | • Generally lower energy consumption compared to thermal desalination methods, especially when using low-temperature waste heat or renewable energy sources. | • Lower energy consumption for low to moderate salinity levels. |
| **Disadvantages** | • Membrane fouling and scaling can occur, requiring regular maintenance. | • High energy consumption.<br>• Requires significant infrastructure and space. | • Requires precise control of temperature and handling of crystals to avoid contamination. Ice separation and melting steps can be challenging to optimize. | • High salinity seawater cannot be used as an input<br>• Membrane fouling can be an issue. |

## Market Insights

- **The global water desalination market size was valued at USD 19.62 billion in 2022. It is estimated to reach USD 43.03 billion by 2031, growing at a CAGR of 9.12%.**
- **In Jan 2024, French renewable energy start-up Sweetch Energy secured a €25 million Series A investment, enabling the first French osmotic power plant to be built in the Rhone delta.**
- **Gulf countries produce 60% of the world's desalinated water. Saudi Arabia and UAE leading these desalination projects with the investments of $14.58 billion and $10.28 billion respectively.**
- **Singapore invested $3billions in desalination over last 5 years, aiming to meet 85% of their future water needs with desalination and NEWater as water consumption now is 440 millions gallon per day and expected to double by 2060.**
- **Taiwan Water Resources Agency has plans for eight new desalination plants across the island. Construction starts in March 2024, with the first plants in Hsinchu and Tainan able to handle 100,000 tons of water per day.**

## Potential Impacts / Outcomes

- **Residents:**

**Water scarcity is a global challenge, affecting over 2 billion people who live in countries where water supply is inadequate. SEAsiders can provide these regions with reliable source of clean freshwater, significantly improving living standards.**

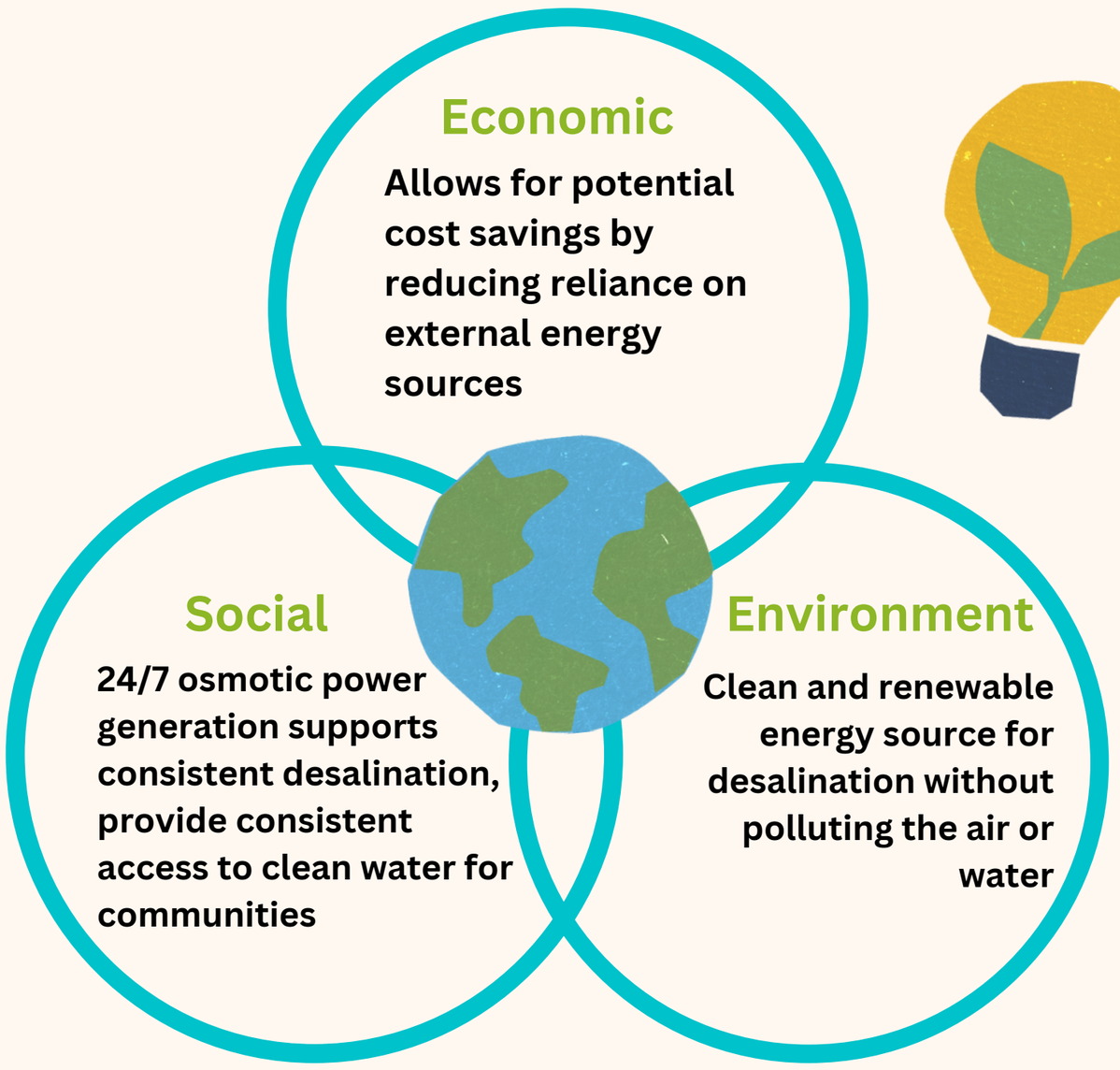
- **Government and Utilities:**

**Traditional desalination methods are expensive and environmentally concerning due to high energy consumption and brine disposal challenges. SEAsiders offers a cost-effective solution. It requires less energy to operate and recycle by-product promoting sustainability and reducing financial burdens.**

- **Industries:**

**Many industries rely heavily on fresh water resources, leading to overexploitation and potential conflicts with local communities. SEAsiders can help by providing a sustainable freshwater supply, reducing industries' dependence on overexploited resources and promoting responsible water management practices.**

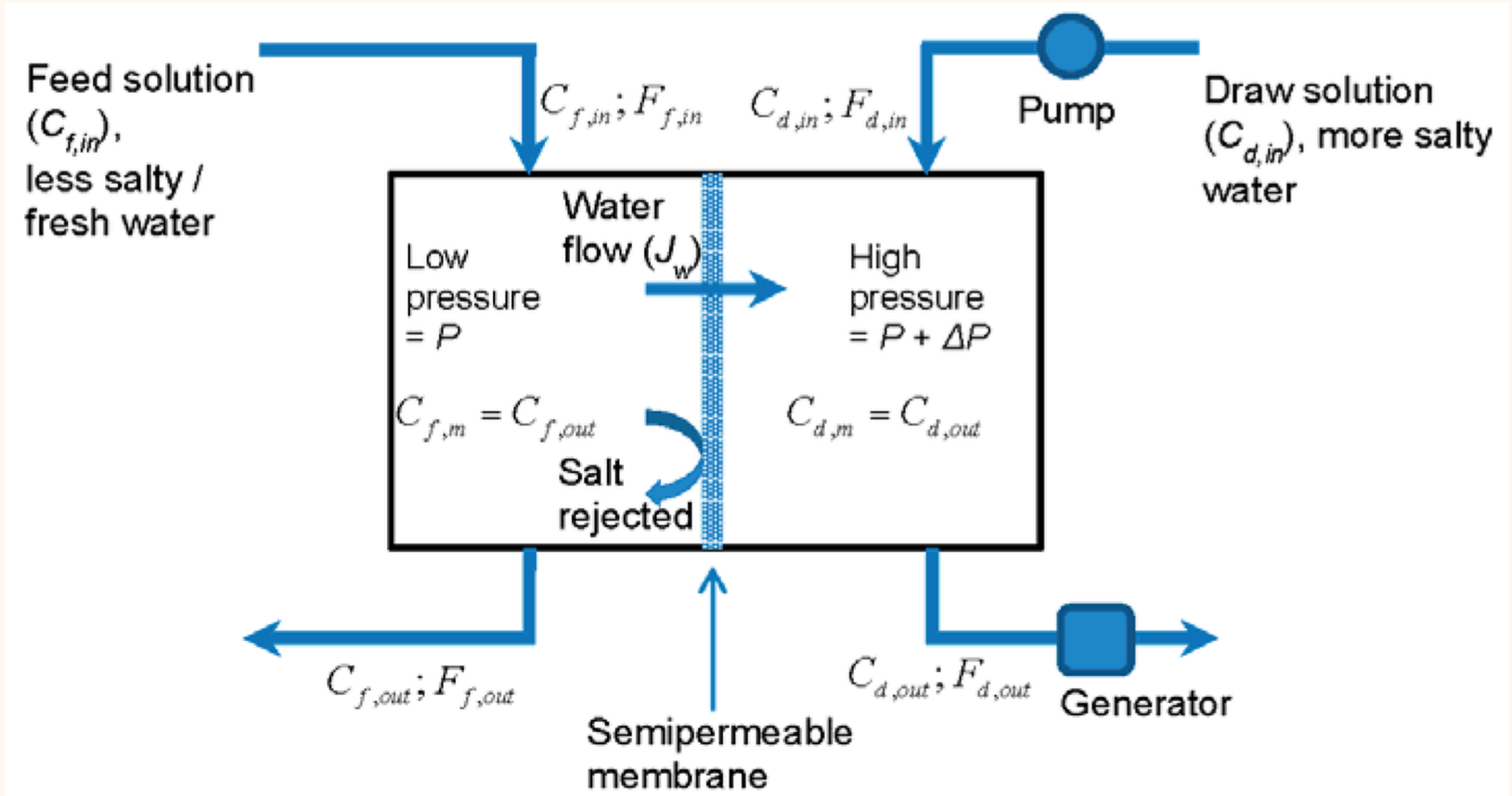
## Sustainability



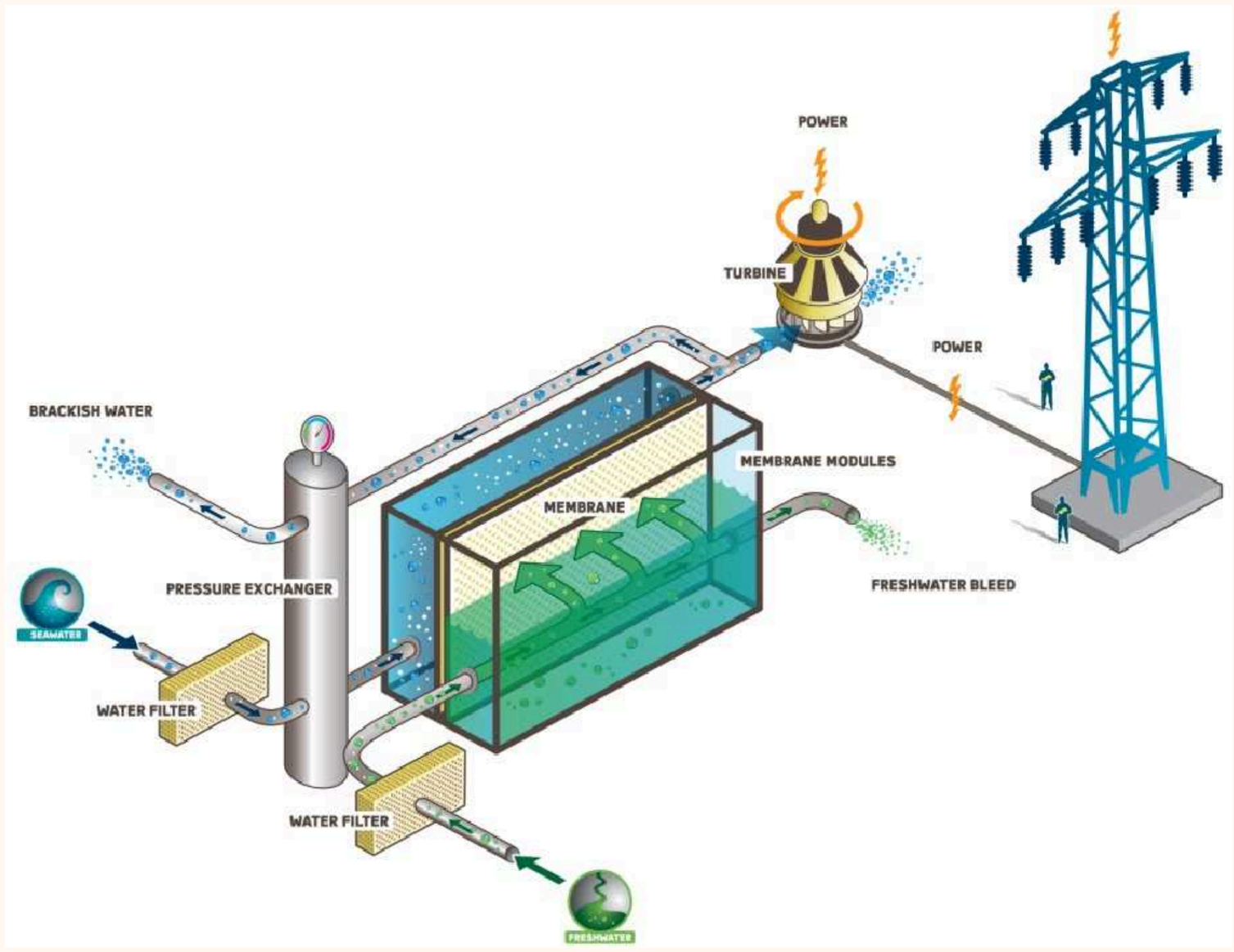
# REFERENCES


1. Touati, K., Dudchenko, A. V., Mauter, M. S., & Rahaman, M. S. (2022). Desalination process design assisted by osmotic power for high water recovery and low energy consumption. ACS Sustainable Chemistry & Engineering, 10(7), 2409–2419. https://doi.org/10.1021/acssuschemeng.1c07078
2. Hsu, J. (2024, January 10). Much of North America may face electricity shortages starting in 2024. New Scientist. https://www.newscientist.com/article/2409679-much-of-north-america-may-face-electricity-shortages-starting-in-2024/
3. The Associated Press. (2023, March 22). 1 in 4 people in the world do not have access to clean drinking water, the U.N. says. NPR. https://www.npr.org/2023/03/22/1165248040/1-in-4-people-in-world-lack-clean-drinking-water-u-n-says
4. Harvey, F. (2023, March 22). Number of city dwellers lacking safe water to double by 2050. The Guardian. https://www.theguardian.com/environment/2023/mar/22/number-city-dwellers-lacking-access-safe-water-double-2050
5. Helfer, F., Lemckert, C., & Anissimov, Y. G. (2014). Osmotic power with Pressure Retarded Osmosis: Theory, performance and trends – A review. Journal of Membrane Science, 453, 337–358. https://doi.org/10.1016/j.memsci.2013.10.053
6. Han, G., Zhang, S., Li, X., & Chung, T.-S. (2015). Progress in pressure retarded osmosis (PRO) membranes for osmotic power generation. Progress in Polymer Science, 51, 1–27. https://doi.org/10.1016/j.progpolymsci.2015.04.005
7. Kurihara, M., & Takeuchi, H. (2018). SWRO-PRO system in “Mega-ton Water System” for energy reduction and low environmental impact. Water (Basel), 10(1), 48-. https://doi.org/10.3390/w10010048
8. Park, K., Kim, J., Yang, D. R., & Hong, S. (2020). Towards a low-energy seawater reverse osmosis desalination plant: A review and theoretical analysis for future directions. Journal of Membrane Science, 595, 117607-. https://doi.org/10.1016/j.memsci.2019.117607
9. Cheng, Z. L., Li, X., & Chung, T.-S. (2018). The forward osmosis-pressure retarded osmosis (FO-PRO) hybrid system: A new process to mitigate membrane fouling for sustainable osmotic power generation. Journal of Membrane Science, 559, 63–74. https://doi.org/10.1016/j.memsci.2018.04.036
10. Alsvik, I. L., & Hägg, M. B. (2013). Pressure retarded osmosis and forward osmosis membranes: Materials and methods. Polymers, 5(1), 303–327. https://doi.org/10.3390/polym5010303
11. Dan Li, Zijing Mo, Anthony G. Fane, Qianhong She, A multifunctional desalination-osmotic energy storage (DOES) system for managing energy and water supply, Desalination, Volume 581,2024,117608, ISSN 0011-9164,https://doi.org/10.1016/j.desal.2024.117608.
12. Yanmei Jiao, Chun Yang, Wenyao Zhang, Qiuwang Wang, Cunlu Zhao, A review on direct osmotic power generation: Mechanism and membranes, Renewable and Sustainable Energy Reviews, Volume 191, 2024, 114078, ISSN 1364-0321, https://doi.org/10.1016/j.rser.2023.114078.
13. Weipeng Chen, Yun Xiang, Xiang-Yu Kong, Liping Wen, Polymer-based membranes for promoting osmotic energy conversion, Giant, Volume 10, 2022, 100094, ISSN 2666-5425, https://doi.org/10.1016/j.giant.2022.100094.
14. Kiho Park, Jungbin Kim, Dae Ryook Yang, Seungkwan Hong, Towards a low-energy seawater reverse osmosis desalination plant: A review and theoretical analysis for future directions, Journal of Membrane Science, Volume 595,2020,117607, ISSN 0376-7388,https://doi.org/10.1016/j.memsci.2019.117607.
15. Menachem Elimelech, William A. Phillip, The Future of Seawater Desalination: Energy, Technology, and the Environment.Science333,712-717(2011).DOI:10.1126/science.1200488

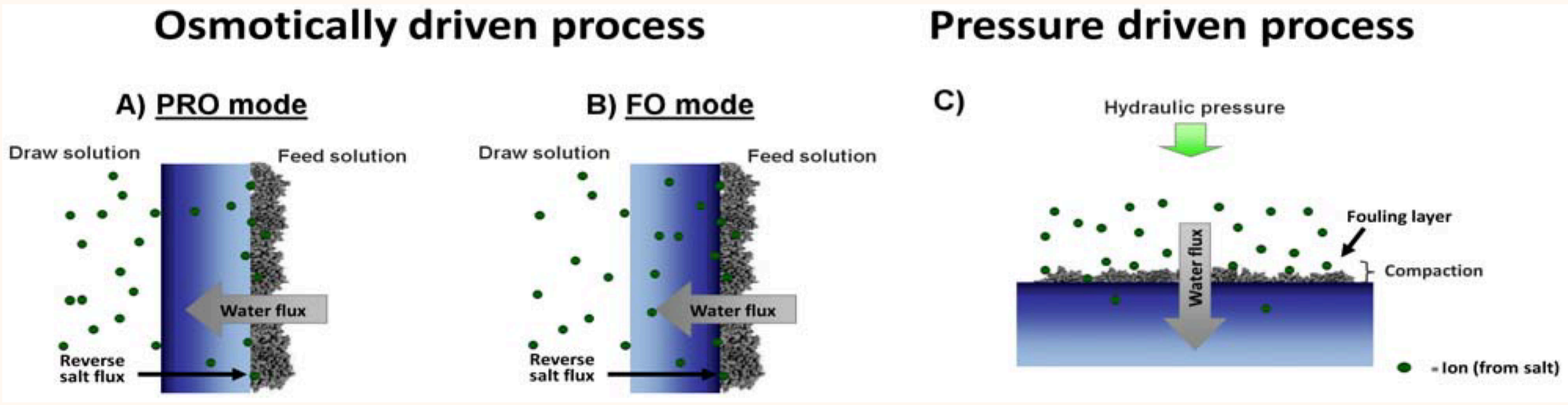
# APPENDIX




**Appendix 1. Model of Pressure Retarded Osmosis**




**Appendix 2. Diagram of commercial PRO osmotic power plant**




**Appendix 3. Illustrations of membrane fouling mechanism in three different osmotic processes**